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Dr Guy Jones
Chief Editor
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Dear Dr Jones,

I am submitting a Data Descriptor entitled **A machine-checked proposition set for digital elevation model analysis** (article type: Data Descriptor; sole author). The Descriptor deposits GeoProofBench v0.1, a versioned catalogue of machine-checkable claims about digital elevation model (DEM) operators. It does not propose a new interpolator. Each concrete claim is archived as a data record: a stable identifier, a one-sentence property, independent Dafny and Lean 4 specifications, numerical gates, rasters or witnesses, and a recorded verdict. Custom code is supplied only to regenerate those verdicts.

The release contains **six** concrete first-order propositions (GPB-001 to 005 and GPB-015; GPB-002b is the two-dimensional companion of 002), **five** composition or counter-example records (P-COMP-1 to 5; P-COMP-1b is the unfilled-ring companion of 1), and **forty-four** stubbed identifiers (GPB-006 to 014 and GPB-016 to 050) reserved for later entries. Stubbed names are not implemented claims.

Independent Dafny and Lean files do not share a kernel. An AutoDL cloud re-run, without the workstation verifier cache, reports 17 verified / 0 errors on `PCOMP_1.dfy`, 12 / 0 on `P006_terminate_under_strict.dfy`, 25 / 0 on `PCOMP_2.dfy`, 20 / 0 on `PCOMP_4_homotopy.dfy`, and 15 / 0 on `PCOMP_5_idempotent.dfy`. These integers are verifier-reported members for each compilation unit (the named file plus its includes), not lemma counts of GeoProofBench as a whole. `PCOMP_1.dfy` itself states 11 lemmas (five for the Fig. 1c chain, plus variants, wrappers, and an example). Lean `lake build` of the deposited VeriGIS library against locked mathlib completes with 0 errors: the library builds. The lockfile imports 2764 mathlib modules; that number is a toolchain size, not a GeoProofBench lemma count, and is not used as evidence that the propositions hold.

Counter-examples are retained as records (`NEGATIVE-RESULT PASS`). An unfilled four-cell ring is a condition-boundary witness: without filling, D8 does not terminate, so P-COMP-1 does not apply. Two small grids show that Zevenbergen–Thorne profile curvature and Horn's second-order slope coefficient are not interchangeable (P-COMP-3). P-COMP-4 records 8 PASS cells and 4 `FAIL-TOLERANCE` cells; all four tolerance cells are 45° resampling and are not relabelled PASS. P-COMP-5 records 12/12 hash agreement across Python, Dafny, and Lean, and 4/4 two-dimensional fill idempotence.

Input rasters include a 5×5 unit plane, a 256^2 synthetic terrain, and two labelled same-size synthetic stand-ins (SRTM-scale 3601^2 and lidar-down 256^2); the stand-ins are not product-accuracy claims. Three public 256^2 windows (USGS 3DEP 1 m lidar, USGS 3DEP 5 m Alaska IFSAR, Copernicus GLO-30) all PASS P-COMP-1. Those products are cited as data: https://doi.org/10.5066/P13LJKFS, https://doi.org/10.5066/P9C064CO, and https://doi.org/10.5270/ESA-c5d3d65.

Exact window parameters are in the article Data Availability section (reviewers do not see this letter).

The GeoProofBench v0.1 dataset is openly available at Science Data Bank, V1: https://doi.org/10.57760/sciencedb.011r9 (CSTR https://cstr.cn/31253.11.sciencedb.011r9). Science Data Bank is the national generalist repository of the Chinese Academy of Sciences; it issues a DOI and a CSTR, and this version is unrestricted under CC-BY-4.0. Software is MIT. Reviewers may download the zip from that DOI without a private link or embargo: Scientific Data is not double-blind, the Descriptor names the author, and the deposit is already open. The laboratory development tree is the public GitHub copy of the same files https://github.com/fariell/verifiable-geocomputation; an anonymous.4open.science mirror is not supplied because it would hide nothing that the article and the DOI do not already disclose.

No previous paper has published this proposition set. A related article (Guo, Liu, Rong, Shi & Tang, *Remote Sens.* **17**, 3662 (2025), https://doi.org/10.3390/rs17223662) concerns hyperspectral semantic segmentation and does not share data with this deposit. Nothing related is under consideration elsewhere. This research received no external funding and has no grant number; it was supported by the author's institutional time. The author declares no competing interests. The work uses synthetic and public elevation rasters only.

Yours sincerely,

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